

Meeting 2 - 10/07/2026

OpenSN & Energy Modelling

OpenSN Architecture

- **OpenSN does not use Mininet.**
- OpenSN uses **Docker containers** for satellites and ground stations.
- Its **Virtual Link Manager** directly manages Linux virtual network devices instead of using Docker's networking layer.
- This reduces networking overhead and improves performance compared with approaches such as StarryNet.
- **VXLAN** is used to connect containers across multiple physical machines.
- OpenSN is designed primarily for **performance and scalability**, rather than detailed physical satellite modelling.

Key OpenSN Design Features

- Separates **configuration** from network/emulation management.
- Uses **etcd / key-value storage** to maintain configuration and system state.
- Uses **JSON configuration**.

Research gap: OpenSN provides a scalable container-based LEO network emulator but does not model satellite energy generation, storage, or energy constraints.

Energy Modelling

Potential model components:

- Solar power generation
- Battery capacity and state of charge
- Satellite power consumption

- Sunlight vs. eclipse periods
- Energy consumed by communication/network activity

Energy state could potentially influence satellite behaviour